Evaluating the Performance and Concordance of a Local SARS-CoV-2 Genomic Surveillance Pipeline

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1. Introduction & Research Problem

- Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is a highly transmissible respiratory pathogen that emerged in late 2019, resulting in the "Coronavirus Disease 2019" (COVID-19) pandemic.
- > 7 million people have succumbed to the COVID-19 disease to date[1].
- **Research Problem:**
  - As viral mutations alter transmissibility, antigenicity, and severity, standardized bioinformatics tools such as PANGOLIN are vital for real-time epidemiological monitoring.
  - However, dependence on high-performance computing (HPC) creates significant barriers for local or resource-constrained settings.

2. Research Question & Objective

Research Question:

- How accurately can a laptop-scale, open-source pipeline reproduce known SARS-CoV-2 reference lineages, and what explains cases of non-concordance?

Objective:

- To evaluate the technical performance, accuracy, reproducibility, and feasibility of executing a laptop-scale public health bioinformatics pipeline for SARS-CoV-2 whole-genome sequencing (WGS) analysis.

3. Project Pipeline

Data Download (SRA Toolkit[2] (fastq-dump --split-files))

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QC & Trimming (fastp[3])

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Reference Alignment (BWA-MEM[4] against Wuhan-Hu-1 (MN908947.3)[5])

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Primer Trimming & Consensus (iVar[6])

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Lineage Assignment (Pangolin[7] + Nextclade[8] CLI)

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Analysis (MultiQC[9] + Python[10] (pandas[11]/matplotlib[12]))

4. Results

accession	reported_lineage	nextclade_call	pangolin_call	nextclade_match	pangolin_match	mean_depth	breadth_covered
SRR11140748	B	B	B	True	True	6306.44	0.9994
SRR11314339	B.1	B.1	B.1	True	True	4177.75	0.9994
SRR11397412	Unassigned	B	Unassigned	False	False	6412.55	0.4353
SRR11397716	B.33	B	B.33	True	True	2682.25	0.9903
SRR11454608	B	B	B	True	True	814.78	1.0
SRR11454612	Unassigned	B	Unassigned	False	False	32.77	0.9769
SRR11454613	B	B	B	True	True	3251.34	1.0
SRR11454614	B	B	B	True	True	6911.52	1.0
SRR11476449	A	B	A	False	True	20221.93	0.9985
SRR13835921	B.1.2	B.1.2	B.1.2	True	True	10058.24	1.0
SRR14294912	P.2	P.2	P.2	True	True	5140.08	0.9999
SRR14294913	P.1	P.1	P.1	True	True	5054.29	0.9997
SRR14460618	B.1.351	B.1.351	B.1.351	True	True	2618.8	0.9993
SRR15891234	AY.37	AY.37	AY.37	True	True	604.84	0.9951
SRR17210511	B.1.240	B.1.240	B.1.240	True	True	2005.13	0.9983
SRR17210512	B.1	B.1	B.1	True	True	3539.6	0.9974

Table 1. Lineage Assignment Concordance and Quality Metrics. Summary of a laptop-scale pipeline evaluating 16 public SARS-CoV-2 paired-end amplicon samples against ground-truth baselines. **Pangolin Accuracy = 87.50%** (14/16), **Nextclade Accuracy = 81.25%** (13/16) . **The ~12–18% discrepancy rate reflects low-coverage amplicon dropout typical in real-world viral surveillance.**

SARS-CoV-2 Lineage Call Concordance v. Sequencing Coverage

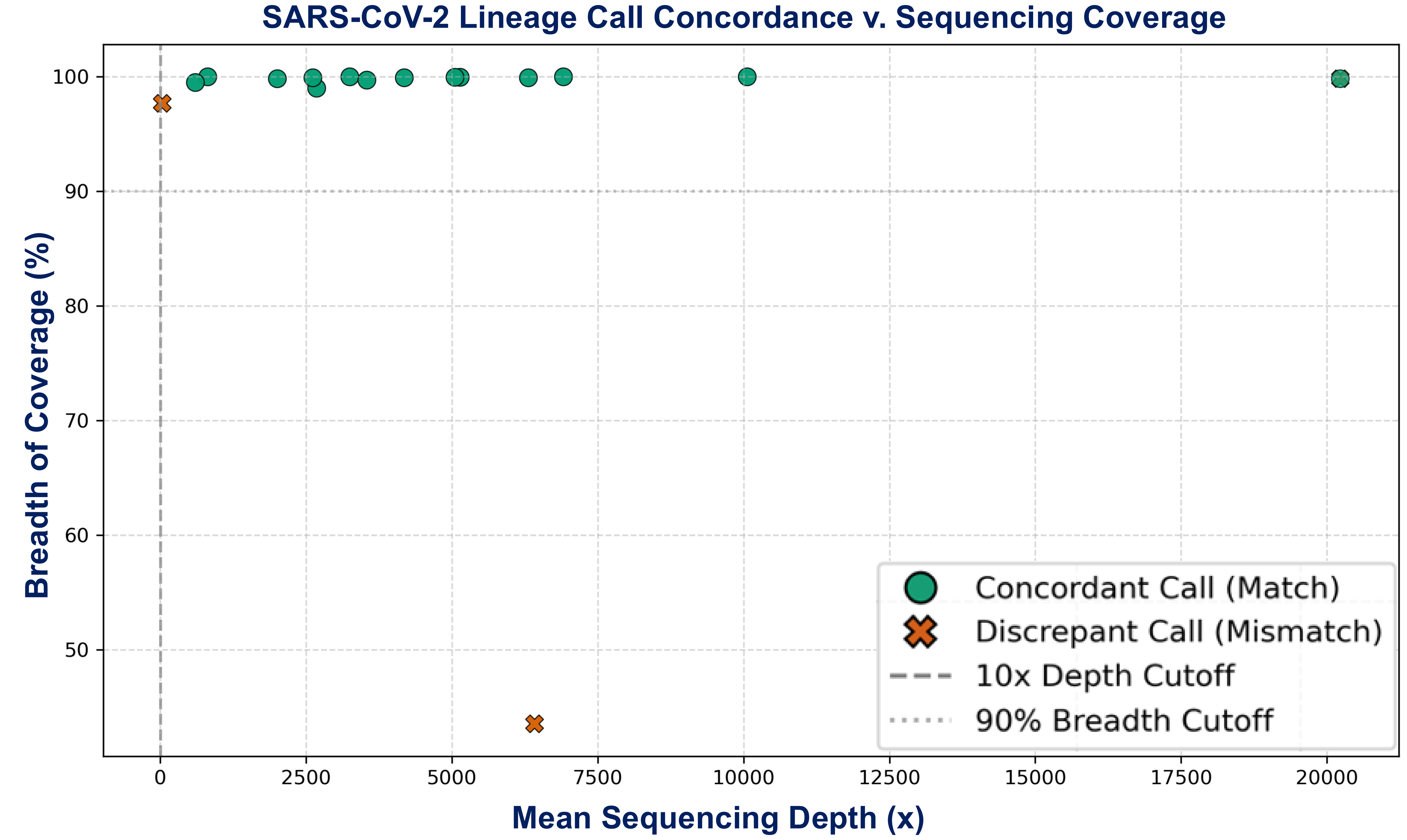

Figure 1. Impact of Sequencing Depth and Coverage Breadth on Lineage Assignment Accuracy. Scatter plot displaying the relationship between mean sequencing depth and genome coverage breadth across 16 public SARS-CoV-2 amplicon samples evaluated with Pangolin and Nextclade CLI. **Vertical dashed line** = 10x mean depth threshold; **Horizontal dotted line** = 90% coverage breadth cut-off.

Root-Cause Analysis

- **SRR11397412 (Amplicon Dropout) =** Low breadth (43.5%) caused N-fills, yielding Pangolin "Unassigned" and Nextclade "B".
- **SRR11454612 (Low Depth) =** Low read coverage caused N-fills that masked critical SNVs, preventing Pangolin and Nextclade from assigning the variant.
- **SRR11476449 (Algorithm Discrepancy/Classification Mismatch) =** Pangolin assigned "A" (UShER tree); Nextclade assigned "B" (mutation distance).

5. Biological Interpretation, Conclusion & Future Work

- **High "True" match rates** confirm the lightweight local pipeline accurately reproduced established lineages.
- **Mismatches between tools** reflect tool-specific algorithmic differences rather than pipeline errors.
- Validating a **laptop-scale** SARS-CoV-2 WGS pipeline has major implications for public health and bioinformatics and decentralizes public health surveillance.
- **Localized variant tracking** enables real-time surveillance in resource-limited settings, speeding outbreak response and driving global sequencing equity.
- **Future pipeline improvements** could include scaling pipeline execution via containerized frameworks (nf-core/viralrecon) and incorporating automated variant annotation (SnpEff/VEP) to streamline functional mutation profiling.

6. References

[1](World Health Organization [WHO], 2026); [2](NCBI, 2026); [3](Chen et al., 2018); [4](Li, 2013); [5](Wu et al., 2020); [6](Grubaugh et al., 2019); [7](O'Toole et al., 2021); [8](Aksamentov et al., 2021); [9](Ewels et al., 2016); [10](Python); [11](Pandas); [12](Hunter,2007)

7. Acknowledgements

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